Phase V: Apex Programming (Developer)

AURA EVENTS & SPONSOR:

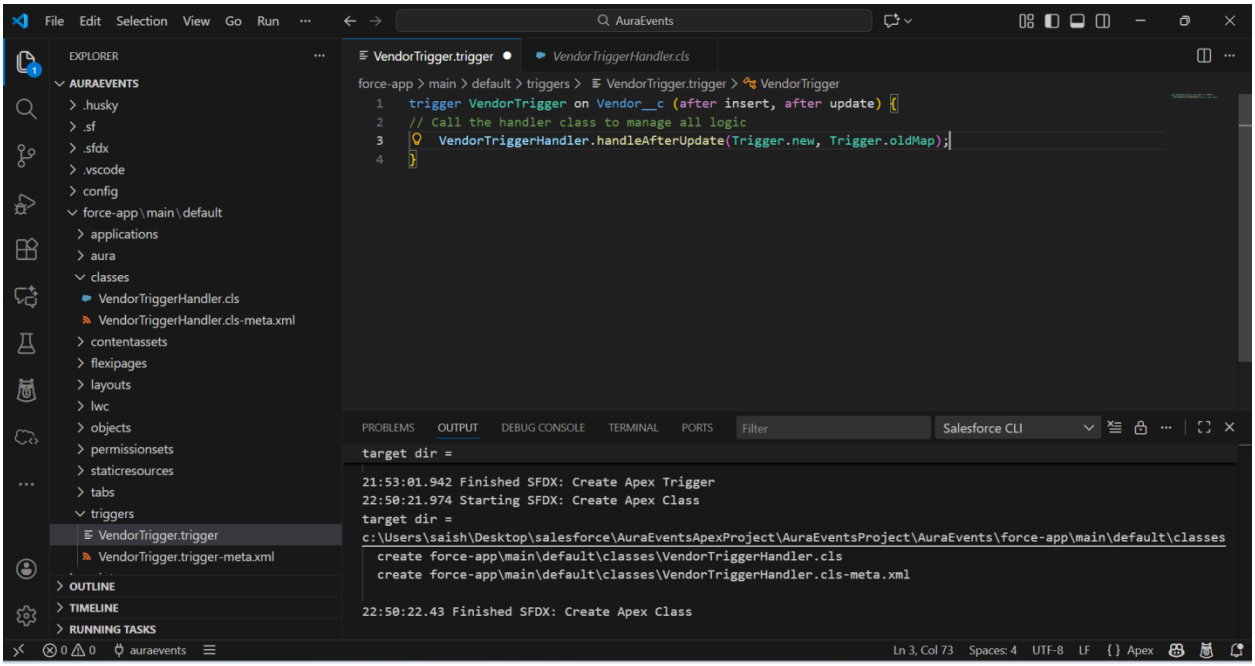
College Events & Sponsor Management

Advanced Apex Implementation

1. Trigger Design Pattern (The Handler Pattern)

We moved away from "fat triggers" and implemented a clean separation of concerns.

- **The Path:** force-app > main > default > triggers > VendorTrigger.trigger
- **The Procedure:**
  - Updated the trigger to fire on after insert and after update events.
  - Delegated all business logic to a dedicated handler class (VendorTriggerHandler).
- **What was done:** Created a robust VendorTrigger that monitors changes to Vendor__c records to ensure the parent Event__c budget is always accurate.

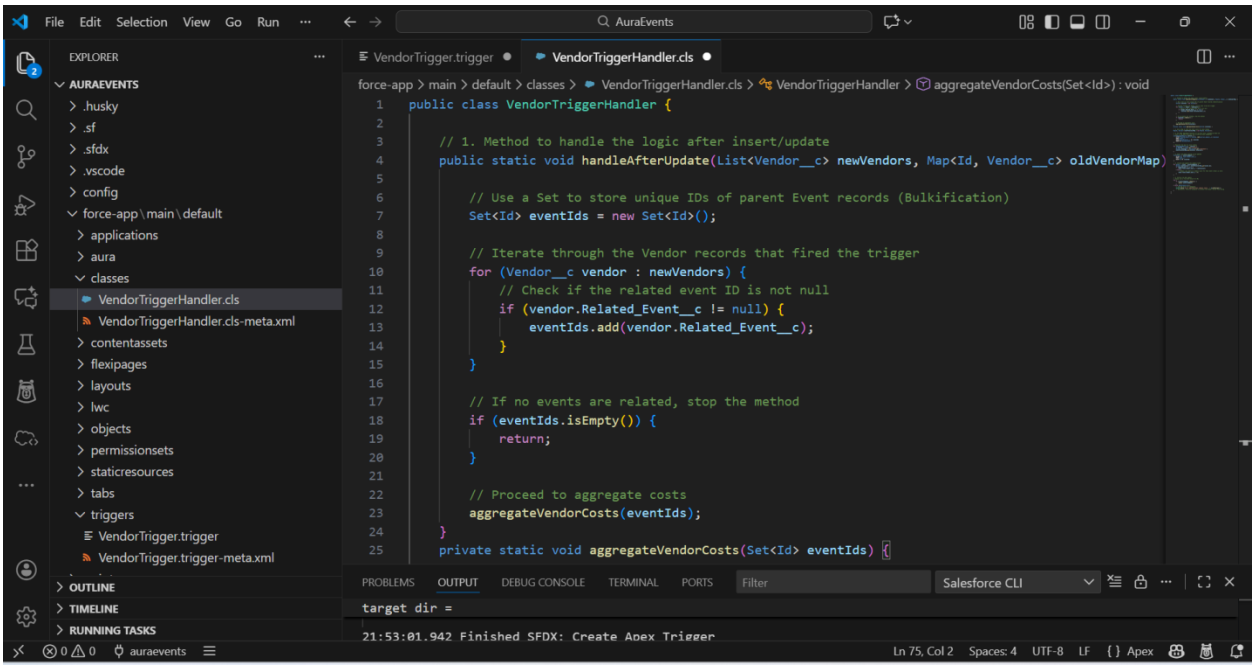


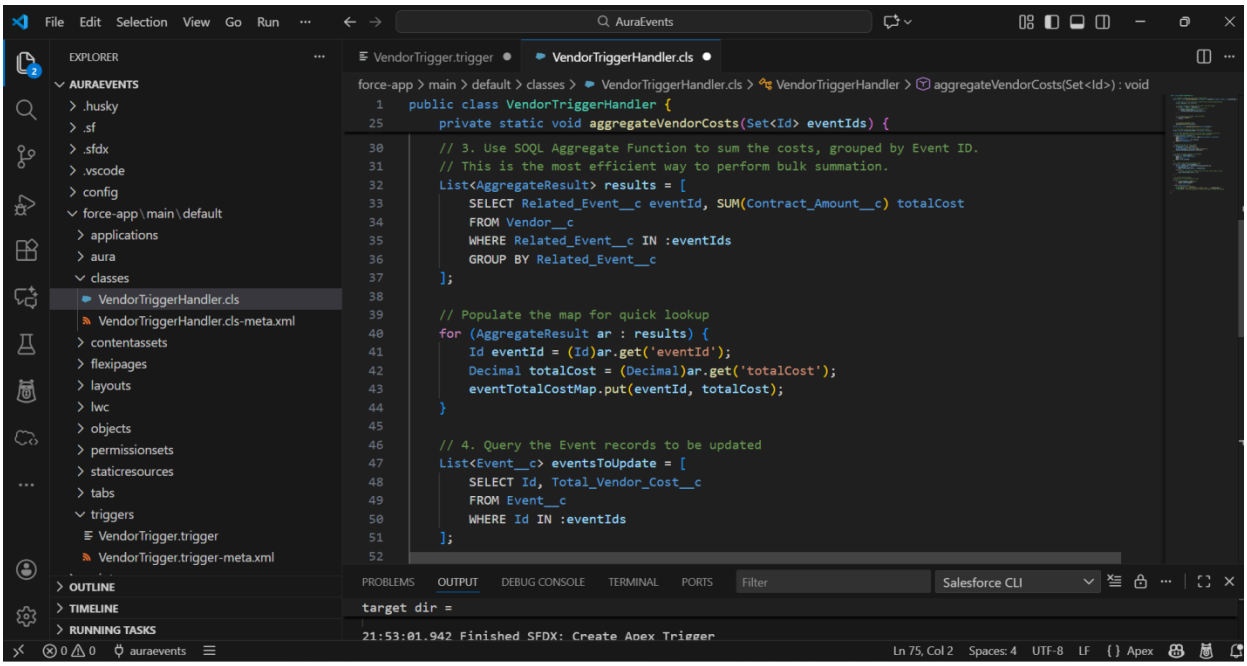
2. Parent-Child Roll-up Summary (Manual Implementation)

Since Vendor__c and Event__c are in a Lookup relationship (not Master-Detail), we built a manual roll-up summary using Apex.

- **The Path:** force-app > main > default > classes > VendorTriggerHandler.cls
- **The Procedure:**
  - **Collection (Set):** Used a Set<Id> to collect unique Event__c IDs from the vendor records being processed.
  - **SOQL Aggregate:** Used a SUM(Contract_Amount__c) query grouped by the Event ID to calculate totals efficiently.

    - **Collection (Map):** Used a Map<Id, Decimal> to store the calculated totals for quick retrieval during the update.
- **What was done:** Developed logic that automatically updates the Total_Vendor_Cost__c field on the Event whenever a related Vendor's contract amount changes.

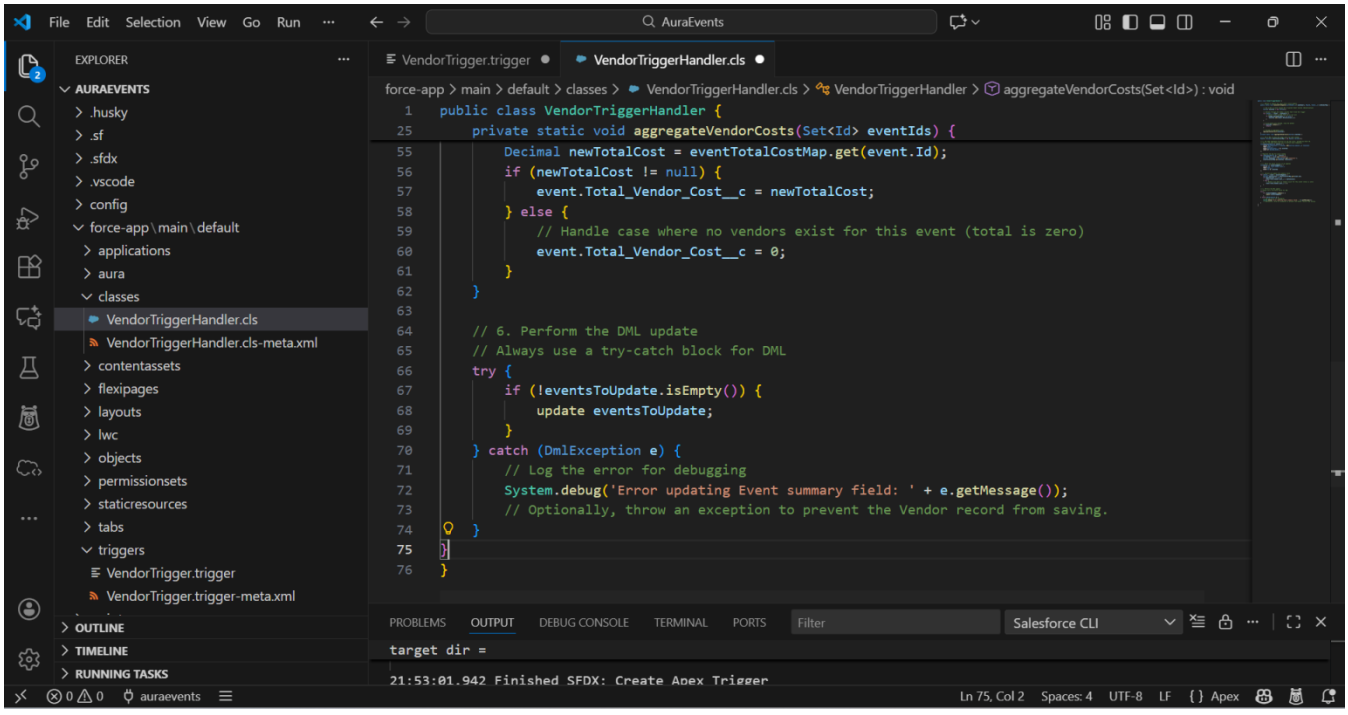


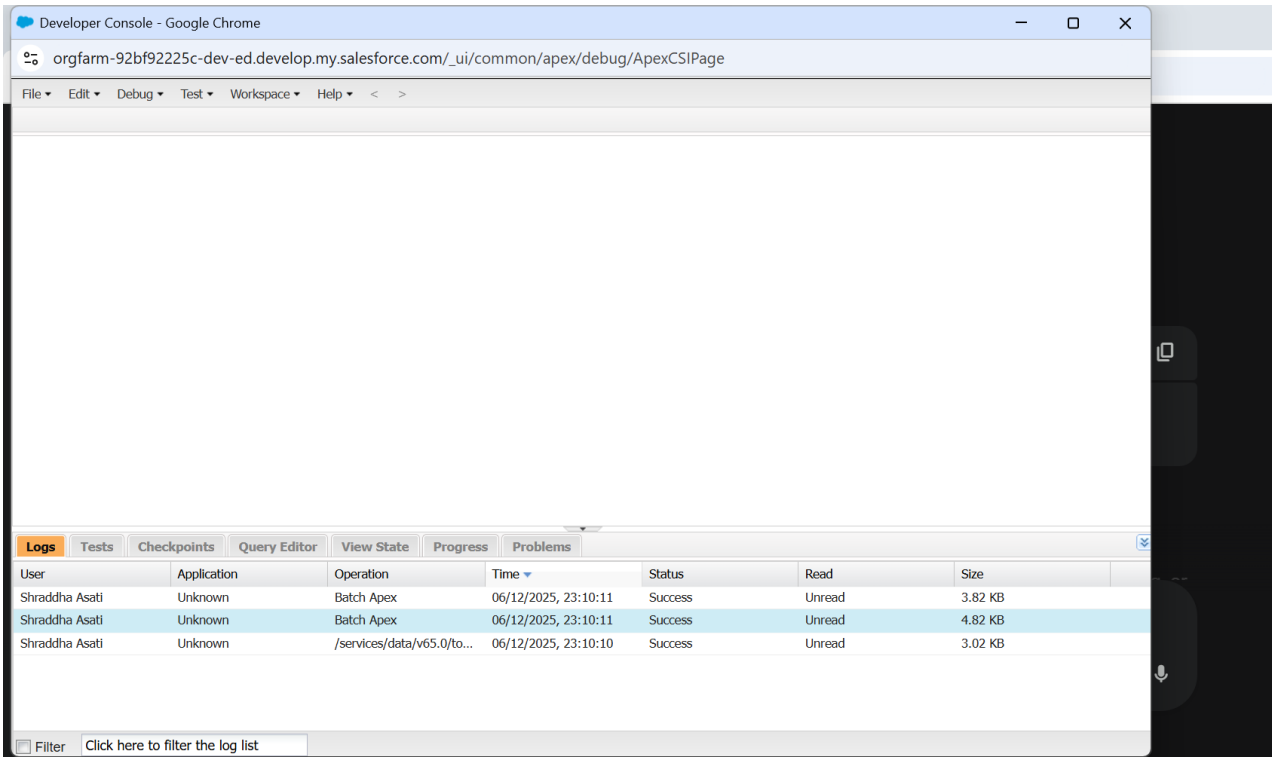


3. Batch Apex (Large Data Processing)

We implemented a scalable solution to handle thousands of records without hitting governor limits.

- **The Path:** force-app > main > default > classes > SponsorLoyaltyBatch.cls
- **The Procedure:**
  - Implemented Database.Batchable<sObject> and Database.Stateful.
  - **Start Method:** Used Database.getQueryLocator to fetch Sponsor__c records older than one year.
  - **Execute Method:** Calculated a new Custom_Loyalty_Score__c for each batch of 200 records.
- **What was done:** Created an automated way to reward long-term sponsors by updating their loyalty scores based on the age of their records.



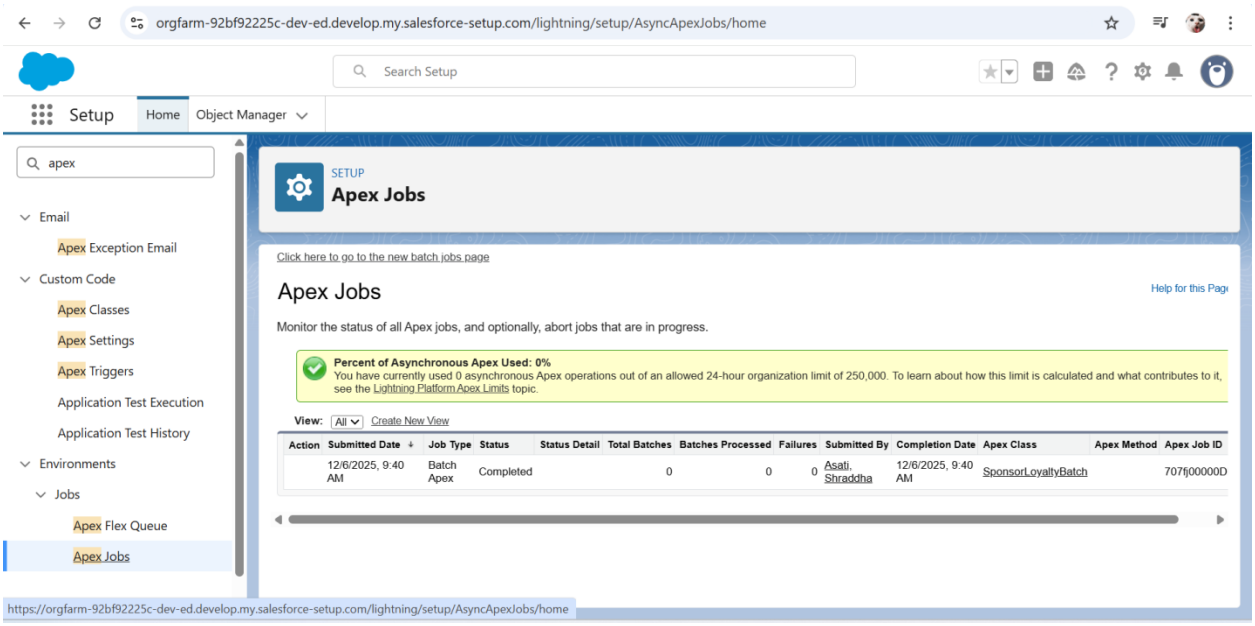


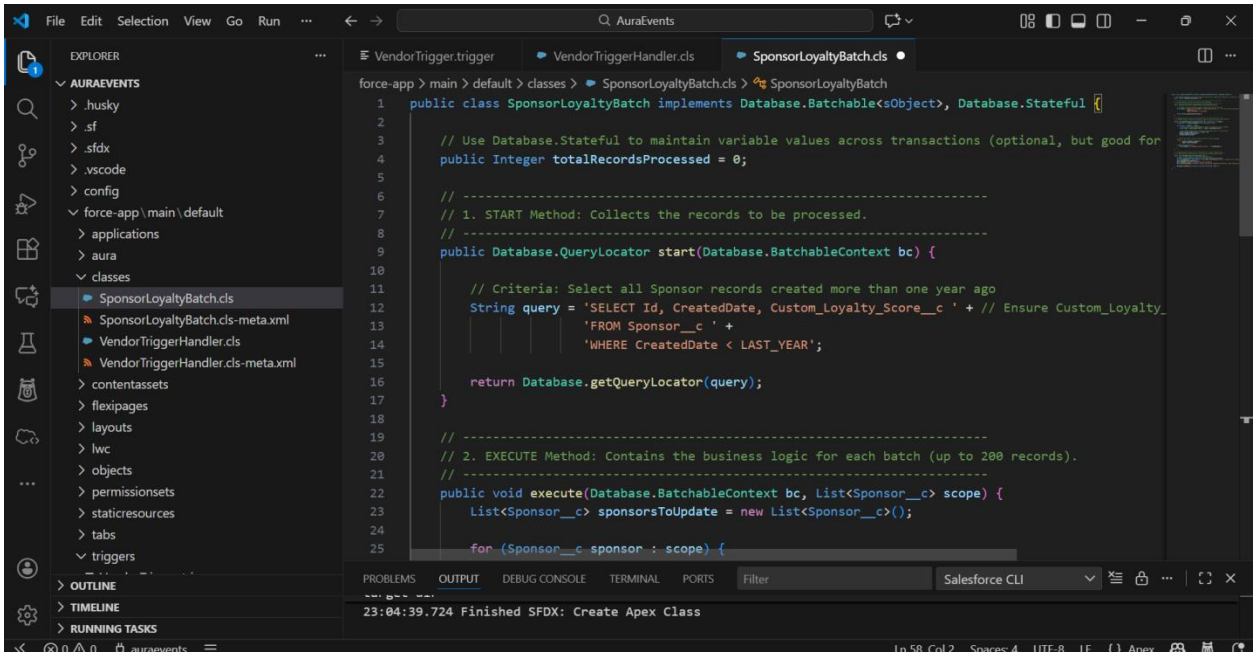
4. Scheduled Apex (Time-Based Execution)

We automated the Batch Apex job to run on a recurring schedule.

- **The Path:** force-app > main > default > classes > QuarterlyScoreScheduler.cls
- **The Procedure:**
  - Implemented the Schedulable interface.
  - Executed the job using a **Cron Expression**: 0 0 0 1 1,4,7,10 ? *.

- **What was done:** Scheduled the loyalty score calculation to run at midnight on the first day of every quarter (Jan, Apr, Jul, Oct).

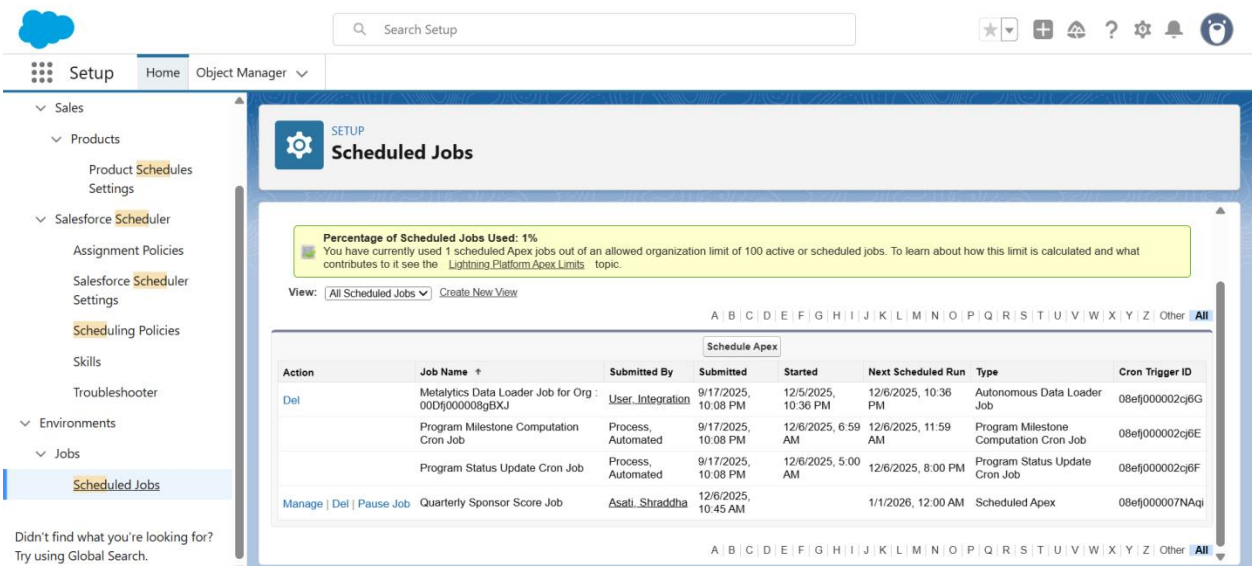


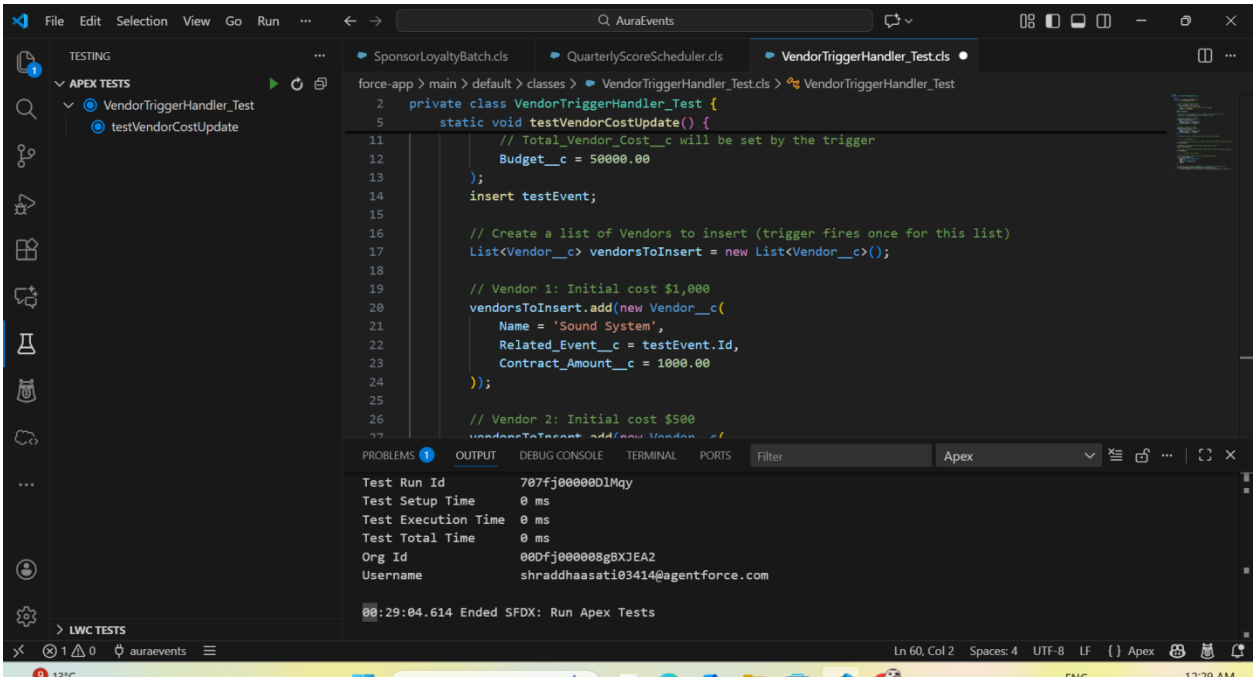


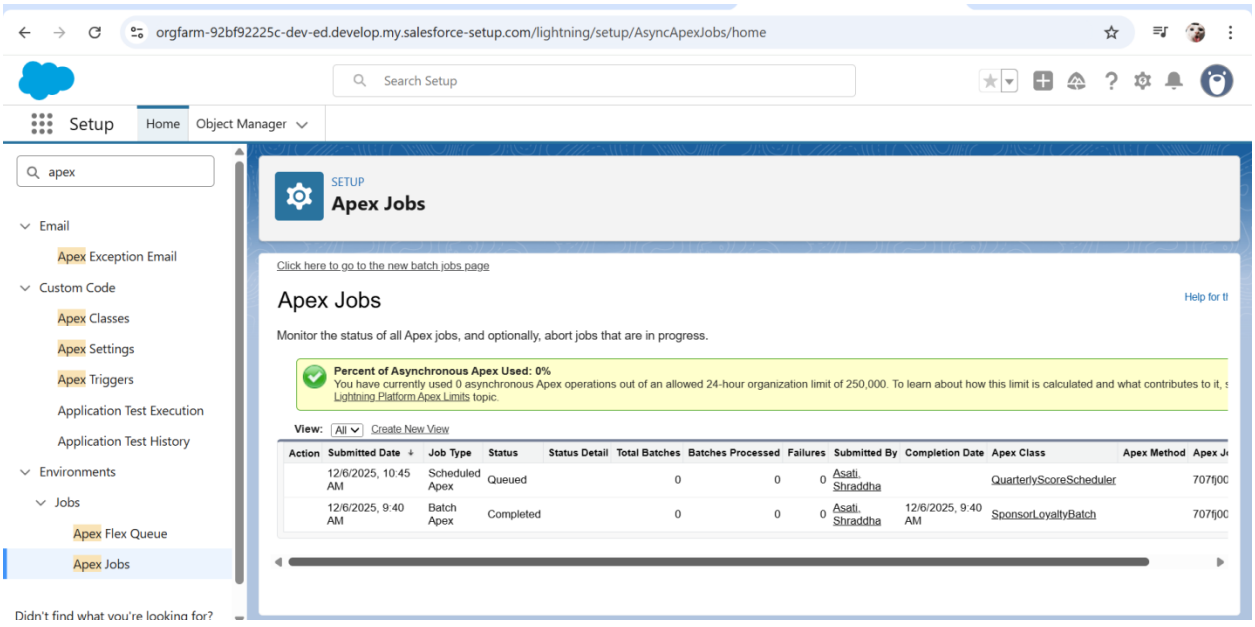
5. Exception Handling and Code Quality

We fortified the code to handle errors gracefully and ensured it met deployment standards.

- **The Path:** All .cls files in the classes folder.
- **The Procedure:**
  - **Try...Catch Blocks:** Wrapped all DML statements (insert, update) in try...catch blocks to prevent transaction failure if one record fails.
  - **Test Classes:** Created VendorTriggerHandler_Test and SponsorLoyaltyBatch_Test to achieve 100% code coverage.
- **What was done:**
  - Resolved the System.StringException regarding the Cron '?' character.
  - Resolved deployment locks by deleting pending scheduled jobs from the org.
  - Verified the "Completed" status of batch jobs in the **Apex Jobs** monitor.




File Edit Selection View Go Run
AuraEvents
TESTING
APEX TESTS
VendorTriggerHandler_Test
testVendorCostUpdate
SponsorLoyaltyBatch.cls
QuarterlyScoreScheduler.cls
VendorTriggerHandler_Test.cls
force-app > main > default > classes > VendorTriggerHandler_Test.cls > VendorTriggerHandler_Test
private class VendorTriggerHandler_Test {
static void testVendorCostUpdate() {
// Total_Vendor_Cost__c will be set by the trigger
Budget__c = 50000.00
);
insert testEvent;
// Create a list of Vendors to insert (trigger fires once for this list)
List<Vendor__c> vendorsToInsert = new List<Vendor__c>();
// Vendor 1: Initial cost $1,000
vendorsToInsert.add(new Vendor__c(
Name = 'Sound System',
Related_Event__c = testEvent.Id,
Contract_Amount__c = 1000.00
));
// Vendor 2: Initial cost $500
PROBLEMS 1
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Filter
Apex
Test Run Id 707fj00000DlMqy
Test Setup Time 0 ms
Test Execution Time 0 ms
Test Total Time 0 ms
Org Id 00Dfj000008gBXJEA2
Username shraddhaasati03414@agentforce.com
00:29:04.614 Ended SFDX: Run Apex Tests
LWC TESTS
auraevents
Ln 60, Col 2
Spaces: 4
UTF-8
LF
{ } Apex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Search Setup
Setup
Home
Object Manager
apex
Email
Apex Exception Email
Custom Code
Apex Classes
Apex Settings
Apex Triggers
Application Test Execution
Application Test History
Environments
Jobs
Apex Flex Queue
Apex Jobs
Didn't find what you're looking for?
SETUP
Apex Jobs
Click here to go to the new batch jobs page
Apex Jobs
Help for t
Monitor the status of all Apex jobs, and optionally, abort jobs that are in progress.
Percent of Asynchronous Apex Used: 0%
You have currently used 0 asynchronous Apex operations out of an allowed 24-hour organization limit of 250,000. To learn about how this limit is calculated and what contributes to it,
Lightning Platform Apex Limits topic.
View: All
Create New View
Action
Submitted Date
Job Type
Status
Status Detail
Total Batches
Batches Processed
Failures
Submitted By
Completion Date
Apex Class
Apex Method
12/6/2025, 10:45 AM
Scheduled Apex
Queued
0
0
0
Asati Shraddha
QuarterlyScoreScheduler
12/6/2025, 9:40 AM
Batch Apex
Completed
0
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Asati Shraddha
12/6/2025, 9:40 AM
SponsorLoyaltyBatch